A proof of the conjectured order of the recurrence for OEIS A230678

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1 The conjecture

The Online Encyclopedia of Integer Sequences records A230678 as

> *Number of 4Xn 0..2 arrays x(i,j) with each element horizontally, diagonally or antidiagonally next to at least one element with value 2-x(i,j)*

and carries in its Formula section the line:

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Empirical recurrence of order 64 (see link above)
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That line carries no separate signature; the entry itself is by R. H. Hardin (Oct 27 2013).

The word *Empirical* — equivalently *Conjecture* — is the entry's own: the statement is recorded as fitted to the published terms, and no proof is given there. As of revision 7, last modified Jun 02 2025, the entry records no proof and links to none, so the statement is open. This note settles it.

The entry states only that the sequence satisfies a linear recurrence of order 64 and refers to a linked file for its coefficients, which this note does not use. What is proved below is the corresponding exact statement: the least order of a linear recurrence the sequence eventually satisfies, and the exact index beyond which it holds.

2 The object

The name is read as follows. The reading is not a guess: it is pinned against the entry's own published terms, and against an independent enumeration, before anything is proved (Section 7).

Let A be an $n' \times 4$ array over $\{0, \dots, 2\}$ with $n' = n$ rows, the entry's array transposed so the growing direction is downwards, the neighbour offsets transposed with it.

The entry's neighbours, *horizontally, diagonally or antidiagonally*, are the cells at the offsets

$$\mathcal{N} = \{(-1,-1), (-1,0), (-1,1), (1,-1), (1,0), (1,1)\}$$

that lie inside the array.

For a cell of value x the entry names the target value $2 - x$. The condition is that at least 1 neighbour of the cell carries that value.

$a(n)$ is the number of such arrays. The entry's offset is 1, so its term of index n is the count for n columns.

3 The count is a walk count

Every neighbour offset moves at most one row, so whether a cell of row i satisfies its condition is decided by rows $i-1$, i and $i+1$. Take as state the last two rows; appending a row decides the row before it, and the accepting condition decides the last one.

An array of n' rows is then exactly a walk of length n' from the empty state to an accepting one, so $a(n) = w^{\mathsf{T}} M^{n'} f$ and the count is C-finite.

4 The degree bound, derived

The reachable part of the digraph has 6643 states, 6643 after trimming; the forward identification of states with equal numbers of completions of every length, then the mirror identification on the edges coming in, leaves $S = 77$ blocks. The count satisfies the linear recurrence given by the characteristic polynomial of that 77×77 matrix; the bound is computed, not assumed.

5 The decision procedure

Let $c_1, \ldots, c_D$ be the coefficients of a claimed recurrence $a(n) = \sum_{i=1}^{D} c_i\, a(n-i)$, let $q(t) = t^D - \sum_{i=1}^{D} c_i t^{D-i}$ be its characteristic polynomial, and put

$$u_m \;=\; \tilde{w}^{\mathsf{T}} L^{m-1} q(L)\, \tilde{f} \qquad (m \geq 1).$$

Expanding the powers of L and using the displayed formula for a, one gets $u_m = a(n) - \sum_{i=1}^{D} c_i\, a(n-i)$ with $n = m + D$. So u_m is exactly the residual of the claim at that index, and the recurrence holds at an index n if and only if the corresponding u_m vanishes.

Now $u_m = \tilde{w}^{\mathsf{T}} L^{m-1} y$ with $y = q(L)\tilde{f}$ a fixed vector, so (u_m) satisfies the characteristic polynomial of L, of degree 77: writing that polynomial as $t^{77} - \sum_{i=1}^{77} \lambda_i t^{77-i}$ gives $u_{m+77} = \sum_{i=1}^{77} \lambda_i u_{m+77-i}$. Hence 77 consecutive vanishing residuals force every later residual to vanish, by induction. The test is therefore finite; and because it locates the *last* nonvanishing residual, it pins the threshold exactly rather than merely bounding it.

Everything is carried out in exact integer arithmetic on the vector $L^{m-1}y$. The matrix M is never formed.

6 Results

Theorem 1. *For A230678, the least order of a linear recurrence that a eventually satisfies is exactly* 64*, and that recurrence holds for every* $n > 65$*, the entry's own figure.*

Proof. The count is C-finite of order at most $S = 77$, so the minimal annihilator of the tail $(a(n))_{n \geq M}$ is the same for every M beyond S: the nilpotent part of L is killed after S steps. Taking such an M and 160 consecutive terms from there, the Berlekamp–Massey algorithm over $\mathbb{Q}$ returns the minimal linear recurrence generating them; since the sequence satisfies one of order at most S, $2S$ terms determine it. Its order is 64. That recurrence was then put through the residual test of Section 5: the last nonvanishing residual is at $n = 65$, followed by more than $S = 77$ consecutive vanishing ones, so it holds for every larger n. No recurrence of smaller order can hold eventually, since the tail's minimal annihilator is what the algorithm returned. $\square$

7 Verification

Four checks were run, each reproducible from the code accompanying this note, and the first two are what pin the reading of the entry's English.

- **The published terms.** The walk count reproduces all 14 terms the entry publishes, exactly, in integer arithmetic, with the index taken from the entry's stated offset (1) rather than fitted. The first of them are 0, 423, 17271, 730503, 37034355, 1844154933, 91258562955, 4522036938501, ...
- **An independent enumeration.** For $n = 1, \ldots, 2$ every one of the 3^{4n} arrays was written out and tested cell by cell against the condition as the entry states it — no transfer matrix, no row window, a separate piece of code. The counts agree with the walk count and with the entry.
- **The claim on the raw data.** Each claim was evaluated on the entry's published terms alone, with no model involved, wherever the proved range and the published data overlap. It holds there.
- **The annihilation test.** Carried to the derived bound $S = 77$ over the whole vector, in exact integer arithmetic, never sampled and never truncated.

8 References

1. OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A230678, revision 7, last modified Jun 02 2025.
2. R. P. Stanley, *Enumerative Combinatorics*, Vol. 1, 2nd ed., Cambridge University Press, 2012; §4.7, the transfer-matrix method.
3. M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011; C-finite sequences and their closure properties.
4. J. Berstel and C. Reutenauer, *Noncommutative Rational Series with Applications*, Cambridge University Press, 2011; linear representations of counting automata and their reduction.